Graph-Based Document Question Answering System — Architecture

INPUT

PDF Document
Input file (any domain, multi-page)
Formats: text, scanned (OCR), tables
pdf_parser.py | PyMuPDF / pdfplumber

User Query
Natural language question (zero-shot)
Languages: English / Hindi / Hinglish
Unicode-normalised | multilingual support

PARSING

PDF Parser
pdf_parser.py
- PyMuPDF Text Extract
- pdfplumber Tables
- OCR Fallback (tesseract)
- Heading Detection
- Text Normalisation
- Section Detection

section_utils.py | sentence_splitter.py

Node Construction
drg_nodes.py | span_extractor.py
- Sentence Nodes
- Span Nodes
- NER Entities
- Section Labels
- Clause / Phrase
- Temporal / Cond.

spaCy sents | regex patterns | NER HRL

Embedding Model
model_cache.py
paraphrase-multilingual-**MiniLM-L12-v2**
- 50+ Language Support
- Shared Cached Instance (CPU)

Query Processing
advanced_retrieval.py
- Unicode NFKC Normalisation
- BM25 Tokenizer
- Query Expander (synonyms)

GRAPH CONSTRUCTION

Document Reasoning Graph (DRG)
drg_graph.py

Node types:
- Sentence Nodes
- TF-IDF Importance

Edge types:
- Adjacent (seq)
- Semantic (cos)
- Entity Coref
- Proximity (+-3)
- Section / Page
- Context (back)

Graph metrics:
- PageRank
- Degree Centrality

DiGraph | directed | adaptive cos threshold (85th pct)

Span Graph
span_graph.py

Span types:
- NER Spans
- Clause / Phrase

Discourse relations:
- Condition
- Requirement
- Temporal
- Negation
- Causation
- Exception

Structural edges:
- Semantic (cos >= 0.7)
- Entity Overlap

DiGraph | directed | adaptive threshold (80th pct)

Graph Analytics
advanced_retrieval.py

Centrality:
- PageRank
- Betweenness
- Closeness
- Eigenvector

Edge weighting:
- Semantic weight
- Structural weight
- Discourse weight
- BM25 Index

Export: graph.json | graph.png

RETRIEVAL

Enhanced Hybrid Reasoner
enhanced_reasoner.py

Retrieval paths:

BM25 Lexical	Semantic Emb.	Graph Traversal	Query Expansion
Inverted index	Cosine sim.	Multi-hop paths	Synonym expand.
TF-IDF IDF	Sentence embed.	Neighbour expan.	Keyword extract.
k1=1.5 b=0.75	Dense retrieval	Directed walk	Multi-variant

Fusion:
- Hybrid Scorer (weighted fusion)
- Centrality Re-ranking (PageRank)

Output:
- Top-k ranked results (sentences + spans, hybrid scores, traversal paths)

Answer Selector
answer_selector.py
- Span De-duplication
- DeBERTa-v3 Extract. QA
- Confidence Estimation
- Evidence Compilation
- Max-length Trimming

QA Evaluator
evaluator.py
- Exact Match (EM)
- Token F1 Score
- Precision / Recall
- Substring Match
- Evidence Recall@K
- Reasoning Depth
- Numeric Comparison
- Multi Ground Truth

HotpotQA | run_custom_qa.py | hotpot_dataset.py

OUTPUT

Answer Output
- Answer Text Span
- Confidence Score
- Evidence Spans (top-5)
- Retrieval Breakdown

Confidence label: High / Medium / Low

Streamlit Web UI (app.py)
- PDF Upload Interface
- Interactive QA Input
- Session State Caching
- Document Statistics

Multilingual query support | Sidebar controls

Graph Visualisation Output
- DRG Graph PNG
- Span Graph PNG
- Span Graph JSON
- Node Importance

Spring layout | edge-type colour coding

Legend
- Input / Answer Output
- Parsing / DRG (sentence)
- Span Graph
- Reasoning / Retrieval
- Evaluation
- Utility / Cache / Viz